



Roebling Labs LLC

We build vessel collision warning systems for vulnerable bridges.

Roebling Labs continuously assigns a threat level to each vessel by combining real-time transponder (AIS) tracking, trajectory forecasting, and AASHTO impact analysis.

Problem

NTSB Recommendation [H-25-030](#) to “incorporate motorist warning systems capable of activating when a threat is identified.”

Solution

The vessel collision warning system by Roebling Labs will alarm if a high-consequence vessel is on a collision course with a vulnerable bridge element. Our system also detects if a vessel collision likely just occurred.

Each vessel is ranked by both consequence and probability of a collision, according with the below risk matrix.

Map of Bridge, Piers, Vessels, and Trajectories

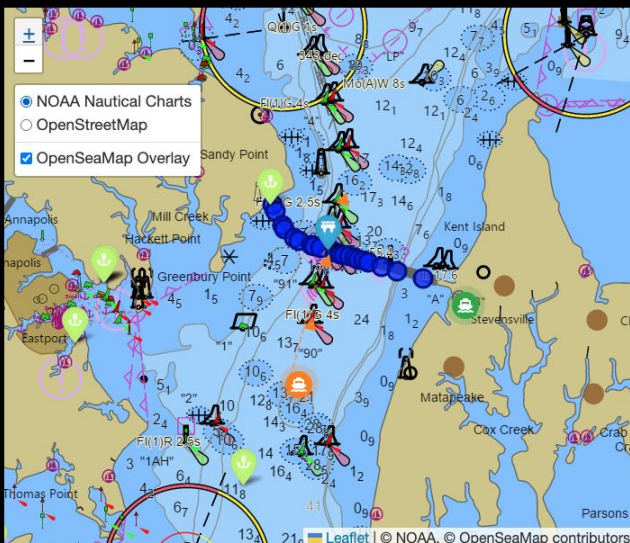


Figure 1 - Screenshot of Dashboard

Prototype Software at: www.vesselcollision.com

Consequence of Collision					
Bridge Ultimate Length (H) / Vessel Impact Force (P) < 10 Calculated per AASHTO	Medium to High	MONITOR (Large vessel moving away from the bridge, far from the bridge, or proceeding on an expected trajectory if approaching the bridge at near range.)	CAUTION - CALL VESSEL (Large vessel approaching the bridge on an unexpected trajectory. Time to attempt contact before closing traffic gates.)	ALARM - CLOSE GATES (Large vessel approaching bridge on a collision course with a pier. Immediate traffic gate closure recommended.)	COLLISION DETECTED - INSPECT FOR DAMAGE (Large vessel likely collided with a bridge pier. Bridge structural damage possible.)
	Low	NEGLIGIBLE THREAT (Anchored, far away, and/or small vessels)	LOW THREAT (Small vessel approaching the bridge at an unexpected trajectory.)	LOW THREAT (Small vessel approaching bridge on collision course with a pier. Bridge unlikely to be damaged or disturbed.)	LOW THREAT (Small vessel likely collided with a bridge pier. Bridge structural damage unlikely.)
		Low	Medium	High	High
		Within 30 nautical miles	Within 2 nautical miles and moving towards the bridge on an unexpected trajectory	Within 1 nautical mile, heading toward bridge, and a trigger of: unexpected trajectory, location, speed, or distress signal	Vessel trajectory stopped, deviated sharply, or terminated at a bridge pier location.
		Probability of Collision			

Figure 2 - Risk Matrix
Vessel threat level updated continuously.

About Us

www.roeblinglabs.com

Roebling Labs LLC is based out of [The Engine](#) “Tough Tech” ecosystem “Built by MIT”.

Bentley Systems, the makers of Microstation CADD software, funded the development of our warning system.



Scott Snelling, P.E.
CEO
20+ Years in bridge industry



Josh Burnett
Chief Technical Officer
20+ years designing electronics and software for life-safety critical medical devices